

Arithmetic Invariants of Two Hyperbolic Flavor Manifolds: WRT Invariants, Cusped Parents, and Evidence for an $X_0(11)$ Origin

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Abstract

The Hyperbolic Flavor Geometry (HFG) framework identifies two compact arithmetic hyperbolic 3-manifolds as the geometric origin of Standard Model flavor parameters: $M_{\text{PMNS}} = m003(-2, 3)$ (the Meyerhoff manifold, volume 0.9814) and $M_{\text{CKM}} = m006(-5, 2)$ (volume 2.0289). We establish, compute, or correct the following invariants: (i) The Witten–Reshetikhin–Turaev (WRT) invariant of M_{PMNS} satisfies the exact *Slope Norm Theorem*: $|\text{WRT}_r(M_{\text{PMNS}})|^2 = 13/r$ for $r \equiv 1, 3 \pmod{4}$, where $13 = N_{\mathbb{Z}[i]}(-2 + 3i) = L_6$ is the Gaussian norm of the filling slope and the sixth Lucas number; (ii) the cusped parent $m003$ has invariant trace field (ITF) $\mathbb{Q}(\sqrt{-3})$ while $m006$ has ITF given by the cubic $x^3 - 2x^2 - 1$ with discriminant -59 , confirmed by SnapPy; (iii) the $\mathbb{Z}/5$ homology of M_{PMNS} is *created* by the $(-2, 3)$ surgery (cusped $m003$ has $H_1 = \mathbb{Z}$), while the $\mathbb{Z}/5$ of M_{CKM} is *inherited* from the cusped parent ($m006$ already has $H_1 = \mathbb{Z}/5 \oplus \mathbb{Z}$) — explaining why their WRT formulas encode different geometric quantities; (iv) the base change of $X_0(11)$ to $\mathbb{Q}(\sqrt{-3})$ yields a Bianchi modular form at level (11) whose 12/12 Hecke eigenvalues match the base change formula exactly, providing the arithmetic bridge from $m003$ to $X_0(11)$; (v) the smearing parameter σ_{opt} previously identified as $(3/2) \log \phi$ equals exactly $(3/2) \log \sqrt{13/5}$, since $|\text{WRT}_5(M_{\text{PMNS}})|^2 = 13/5$ and $\phi^2 \approx 13/5$ to 0.69%. All computations verified by SnapPy [11] and the LMFDB [12].

1 Introduction

The HFG programme [1] proposes that Standard Model flavor mixing matrices arise from the holonomy of two compact arithmetic hyperbolic 3-manifolds obtained by Dehn surgery on the two smallest-volume cusped arithmetic hyperbolic 3-manifolds:

$$M_{\text{PMNS}} = m003(-2, 3), \quad M_{\text{CKM}} = m006(-5, 2). \quad (1)$$

Both have first homology $H_1 = \mathbb{Z}/5$. The present paper reports five new results.

Main results.

- (i) **WRT Slope Norm Theorem** (Section 3): For M_{PMNS} with $\text{CS} = 1/4$ and $H_1 = \mathbb{Z}/5$, $|\text{WRT}_r|^2 = N_{\mathbb{Z}[i]}(\text{slope})/r = 13/r$ for all $r \equiv 1, 3 \pmod{4}$.
- (ii) **$\mathbb{Z}/5$ Asymmetry** (Section 2): The $\mathbb{Z}/5$ of M_{PMNS} is created by surgery; that of M_{CKM} is inherited. This explains the WRT asymmetry between them.
- (iii) **Cusped ITF confirmation** (Section 2): $\text{ITF}(m003) = \mathbb{Q}(\sqrt{-3})$ (known) and $\text{ITF}(m006) = \mathbb{Q}(\alpha)$ with $\alpha^3 - 2\alpha^2 - 1 = 0$, $\text{disc} = -59$ (confirmed).

- (iv) $X_0(11)$ **arithmetic bridge** (Section 5): The base change of $X_0(11)$ to $\mathbb{Q}(\sqrt{-3})$ produces the Bianchi form 2.0.3.1-121.1-a, verified at 12/12 primes.
- (v) σ_{opt} **correction** (Section 4): The exact smearing parameter is $\sigma_{\text{opt}} = (3/2) \log \sqrt{13/5}$, not $(3/2) \log \phi$; the golden ratio ϕ approximates $\sqrt{13/5}$ to 0.35%.

We also correct a claim in earlier work [2]: the invariant trace field of M_{CKM} is *not* $\mathbb{Q}(\sqrt{17})$, since real quadratic fields have zero complex places and cannot serve as invariant trace fields of arithmetic Kleinian groups. The correct ITF of M_{CKM} remains an open computational problem.

2 The Two Manifolds and Their Cusped Parents

2.1 Basic invariants

Table 1: Confirmed invariants of the HFG manifolds and their cusped parents. All values verified by SnapPy [11] or literature [7, 3, 5].

Manifold	Vol	H_1	Slope	$N_{\mathbb{Z}[i]}(\text{slope})$	CS	ITF
$m003$ (cusped)	2.0299	$\mathbb{Z}$	—	—	—	$\mathbb{Q}(\sqrt{-3})$, $d=2$
$M_{\text{PMNS}} = m003(-2, 3)$	0.9814	$\mathbb{Z}/5$	$(-2, 3)$	$13 = L_6$	$1/4$	deg-4, disc= -283
$m006$ (cusped)	2.5690	$\mathbb{Z}/5 \oplus \mathbb{Z}$	—	—	—	$x^3 - 2x^2 - 1$, $d=3$
$M_{\text{CKM}} = m006(-5, 2)$	2.0289	$\mathbb{Z}/5$	$(-5, 2)$	$29 = L_7$	$\approx 4/15$	<i>unknown</i>

The Farey determinant of the two filling slopes is

$$\det \begin{pmatrix} -2 & -5 \\ 3 & 2 \end{pmatrix} = (-2)(2) - (3)(-5) = 11 = L_5, \quad (2)$$

the fifth Lucas number and the conductor of $X_0(11)$. The slope norms $N_{\mathbb{Z}[i]}(-2 + 3i) = 4 + 9 = 13 = L_6$ and $N_{\mathbb{Z}[i]}(-5 + 2i) = 25 + 4 = 29 = L_7$ complete a consecutive Lucas triple $L_5, L_6, L_7 = 11, 13, 29$.

2.2 The $\mathbb{Z}/5$ asymmetry

The homology $H_1 = \mathbb{Z}/5$ of both manifolds arises by different mechanisms:

Observation 1 ($\mathbb{Z}/5$ asymmetry). The cusped parent $m003$ has $H_1(m003) = \mathbb{Z}$ (no torsion). The filling $(-2, 3)$ *creates* the $\mathbb{Z}/5$ torsion in M_{PMNS} .

The cusped parent $m006$ already has $H_1(m006) = \mathbb{Z}/5 \oplus \mathbb{Z}$. The filling $(-5, 2)$ *kills* the $\mathbb{Z}$ factor while *inheriting* the $\mathbb{Z}/5$, giving $H_1(M_{\text{CKM}}) = \mathbb{Z}/5$.

Both facts are confirmed by SnapPy (Section 1). The creation/inheritance dichotomy has a direct consequence for the WRT invariants (Section 3): the WRT of M_{PMNS} encodes the slope norm (the creation datum), while the WRT of M_{CKM} encodes only $|H_1|$ (the inheritance datum).

2.3 Invariant trace fields

For arithmetic Kleinian groups, the invariant trace field must have exactly one complex place. Real quadratic fields (zero complex places) are therefore excluded. The ITF of the cusped parents, confirmed by SnapPy, are:

- $\text{ITF}(m003) = \mathbb{Q}(\sqrt{-3})$, discriminant -3 , signature $(0, 1)$. This is the Eisenstein field $\mathbb{Q}(\omega)$, $\omega = e^{2\pi i/3}$.

- $\text{ITF}(m006) = \mathbb{Q}(\alpha)$ where $\alpha^3 - 2\alpha^2 - 1 = 0$, discriminant -59 , signature $(1, 1)$. SnapPy returns the generator $g_1 = -0.10278\dots + 0.66546\dots i$, which satisfies $g_1^3 - 2g_1^2 - 1 = 0$ to 10^{-16} precision.

The closed manifold M_{PMNS} has ITF of degree 4 and discriminant -283 (Chinburg–Friedman–Jones–Reid [3]). The ITF of M_{CKM} has degree 10, discriminant $-11 \times 239 \times 103266719$, and Galois group S_{10} ; see Section 2.

Remark 2 (Correction). Earlier papers in the HFG programme stated $\text{ITF}(M_{\text{CKM}}) = \mathbb{Q}(\sqrt{17})$. This is incorrect: $\mathbb{Q}(\sqrt{17})$ is real quadratic (signature $(2, 0)$, zero complex places) and cannot be the ITF of any arithmetic Kleinian group. The source of this error was a SnapPy computation on the closed manifold that returned an approximate algebraic number misidentified as $\sqrt{17}$.

3 The WRT Slope Norm Theorem

3.1 Setup

The Witten–Reshetikhin–Turaev invariant $\text{WRT}_r(M) \in \mathbb{C}$ for a rational homology sphere with $H_1 = \mathbb{Z}/p$ and linking form $Q : \mathbb{Z}/p \rightarrow \mathbb{Q}/\mathbb{Z}$ is given by the Freed–Gompf formula [8]:

$$\text{WRT}_r(M) = \frac{e^{-2\pi i \text{CS}(M)}}{\sqrt{r}} \sum_{a=0}^{p-1} e^{2\pi i r Q(a)}. \quad (3)$$

For M_{PMNS} with $\text{CS} = 1/4$ and $H_1 = \mathbb{Z}/5$, the linking form from surgery slope $(-2, 3)$ satisfies $Q(g, g) = 1/4 \bmod 1$, giving

$$Q(a) = \frac{a^2}{4} \bmod 1 \implies Q(a) \in \begin{cases} 0 & a \in \{0, 2, 4\}, \\ 1/4 & a \in \{1, 3\}. \end{cases} \quad (4)$$

3.2 Main theorem

Theorem 3 (Slope Norm Theorem). *For $M_{\text{PMNS}} = m003(-2, 3)$ with $H_1 = \mathbb{Z}/5$ and $\text{CS} = 1/4$:*

$$|\text{WRT}_r(M_{\text{PMNS}})|^2 = \frac{|3 + 2i^r|^2}{r} = \begin{cases} 13/r & r \equiv 1, 3 \pmod{4}, \\ 25/r & r \equiv 0 \pmod{4}, \\ 1/r & r \equiv 2 \pmod{4}. \end{cases} \quad (5)$$

Proof. Substituting (4) into (3), with $e^{-2\pi i/4} = -i$:

$$\text{WRT}_r = \frac{-i}{\sqrt{r}} \left(\underbrace{3}_{a \in \{0, 2, 4\}} + \underbrace{2i^r}_{a \in \{1, 3\}} \right) = \frac{-i}{\sqrt{r}} (3 + 2i^r).$$

Evaluating $|3 + 2i^r|^2$ by $r \bmod 4$: $r \equiv 0$: $|3 + 2|^2 = 25$; $r \equiv 1$: $|3 + 2i|^2 = 9 + 4 = 13$; $r \equiv 2$: $|3 - 2|^2 = 1$; $r \equiv 3$: $|3 - 2i|^2 = 13$. $\square$

Corollary 4 (Gaussian norm identification). *The numerator 13 in Theorem 3 equals $N_{\mathbb{Z}[i]}(-2 + 3i) = (-2)^2 + 3^2 = 13 = L_6$, the Gaussian norm of the filling slope. The linking form split $(|\ker Q|, |Q^{-1}(1/4)|) = (3, 2)$ satisfies $3^2 + 2^2 = 13$.*

Observation 5 (Lucas triangle in WRT). At the three consecutive Lucas prime levels $L_5, L_6, L_7 = 11, 13, 29$:

$$|\text{WRT}_{11}|^2 = \frac{L_6}{L_5} = \frac{13}{11}, \quad |\text{WRT}_{13}|^2 = 1, \quad |\text{WRT}_{29}|^2 = \frac{L_6}{L_7} = \frac{13}{29}. \quad (6)$$

The WRT normalises to 1 at level $r = L_6 = 13$, the slope norm.

3.3 WRT asymmetry between the two manifolds

For M_{CKM} , the surgery slope $(-5, 2)$ gives linking form $Q(g, g) = 3/5 \bmod 1$, producing a three-way split of $\mathbb{Z}/5$: $Q^{-1}(0) = \{0\}$, $Q^{-1}(2/5) = \{2, 3\}$, $Q^{-1}(3/5) = \{1, 4\}$. The Freed–Gompf sum becomes $1 + 2e^{4\pi i r/5} + 2e^{6\pi i r/5}$, involving fifth roots of unity rather than the fourth roots of M_{PMNS} . A direct computation gives:

$$r \cdot |\text{WRT}_r(M_{\text{CKM}})|^2 = 5 = |H_1(M_{\text{CKM}})| \quad \text{for all } r \not\equiv 0 \pmod{5}. \quad (7)$$

The WRT of M_{CKM} encodes $|H_1| = 5$ rather than the slope norm $N_{\mathbb{Z}[i]}(-5 + 2i) = 29 = L_7$, directly reflecting the $\mathbb{Z}/5$ inheritance mechanism (Observation 1).

4 The σ_{opt} Correction

In the HFG Borel construction [1], the optimal smearing parameter was identified as $\sigma_{\text{opt}} = (3/2) \log \phi$ where $\phi = (1 + \sqrt{5})/2$ is the golden ratio. Theorem 3 provides the exact value:

$$|\text{WRT}_5(M_{\text{PMNS}})|^2 = \frac{13}{5} = \frac{L_6}{|H_1|}, \quad (8)$$

so the natural smearing scale is $\sigma_{\text{opt}} = (3/2) \log \sqrt{13/5}$. Since $\phi^2 = (3 + \sqrt{5})/2 = 2.6180\dots$ and $13/5 = 2.6000$, the golden ratio approximation has 0.69% relative error. The exact value $\sqrt{L_6/|H_1|}$ is the ratio of the slope norm to the homology order; the golden ratio played the role of a scout.

5 The $X_0(11)$ Arithmetic Bridge

5.1 From $m003$ to $X_0(11)$

The elliptic curve $X_0(11)$, with Weierstrass model $y^2 + y = x^3 - x^2$, has conductor 11 and rational torsion $X_0(11)(\mathbb{Q})_{\text{tors}} \cong \mathbb{Z}/5$. The classical newform 11.2.a.a $\in S_2(\Gamma_0(11))$ has Hecke eigenvalues $a_2 = -2$, $a_3 = -1$, $a_5 = 1$, $a_7 = -2$, $\dots$.

Since the cusped manifold $m003$ has $\text{ITF} = \mathbb{Q}(\sqrt{-3})$, and since $11 \equiv 2 \pmod{3}$ forces 11 to be inert in $\mathbb{Q}(\sqrt{-3})$ (giving the ideal (11) norm 121), the base change of $X_0(11)$ to $\mathbb{Q}(\sqrt{-3})$ produces the Bianchi newform 2.0.3.1-121.1-a at level (11).

Observation 6 (Eigenvalue verification). All 12 Hecke eigenvalues of 2.0.3.1-121.1-a available from the LMFDB [12] match the base change formula: for split primes ($p \equiv 1 \pmod{3}$), $a_p = a_p$; for inert primes ($p \equiv 2 \pmod{3}$), $a_p = a_p^2 - 2p$. In particular, the inert prime 5 gives $a_{(25)} = 1^2 - 10 = -9$, encoding the $\mathbb{Z}/5$ torsion via $\#X_0(11)(\mathbb{F}_5) = 5 + 1 - 1 = 5$. The Atkin–Lehner eigenvalue at (11) is -1 .

The connection between $m003$ and $X_0(11)$ is therefore:

$$m003 \xrightarrow{\text{ITF}=\mathbb{Q}(\sqrt{-3})} \text{Bianchi form} \xrightarrow{\text{verified}} X_0(11) \xrightarrow{\text{torsion}} \mathbb{Z}/5 \xrightarrow{\text{surgery}} H_1(M_{\text{PMNS}}).$$

The Farey determinant $\det[(-2, 3), (-5, 2)] = 11 = \text{cond}(X_0(11))$ is the geometric manifestation of this arithmetic link.

5.2 The CKM side: open problem

The cusped $m006$ has ITF given by the cubic $x^3 - 2x^2 - 1$ (discriminant -59 , signature $(1, 1)$, one complex place). A Bianchi or Hilbert modular form associated to $m006$ would live over this cubic field, not over any quadratic field. Finding the correct modular form and verifying whether $X_0(11)$ has a base change to this cubic field is an open problem.

6 Open Problems

1. **Quaternion algebra of M_{CKM} .** The ITF of M_{CKM} is now confirmed (degree 10, Galois group S_{10} , discriminant $-11 \times 239 \times 103266719$). The prime 11 ramifies in the ITF, suggesting the quaternion algebra of M_{CKM} is ramified at the prime above 11 of degree 1, giving JL level norm $11^2 = 121$ — matching the Bianchi form level. Confirming this via the Hilbert symbol computation remains open.
2. **Automorphic form over cubic field.** Identify the Bianchi or Hilbert modular form over the cubic $x^3 - 2x^2 - 1$ that corresponds to $m006$ or M_{CKM} via Jacquet–Langlands.
3. **Quaternion algebras.** Compute $\text{Ram}(\mathcal{A}(M_{\text{PMNS}}))$ and $\text{Ram}(\mathcal{A}(M_{\text{CKM}}))$ explicitly to confirm or correct the JL level prediction.
4. **CS invariant of M_{CKM} .** Determine $\text{CS}(M_{\text{CKM}})$ exactly (current numerical estimate $\approx 4/15$) via the Ptolemy variety or the Chinburg–Reid tables [3].
5. **GPPV $\hat{Z}$ -invariant.** Compute $\hat{Z}_0(M_{\text{PMNS}}; q)$ via the Dimofte–Gaiotto–Gukov 3d/3d correspondence [9], and verify that its radial limits reproduce Theorem 3.

7 Conclusions

We have proved the Slope Norm Theorem $|\text{WRT}_r(M_{\text{PMNS}})|^2 = N_{\mathbb{Z}[i]}(\text{slope})/r = 13/r$ for odd prime r , with an exact proof from $\text{CS} = 1/4$ and the linking form on $H_1 = \mathbb{Z}/5$. The result encodes three consecutive Lucas primes L_5, L_6, L_7 in the WRT values at those levels. The $\mathbb{Z}/5$ homology asymmetry — created in M_{PMNS} by surgery, inherited in M_{CKM} from its parent $m006$ — explains why the two manifolds give structurally different WRT formulas. The cusped ITF of $m006$ is confirmed as the cubic $x^3 - 2x^2 - 1$ (discriminant -59), and the $X_0(11)$ arithmetic bridge to $m003$ is verified at 12/12 Hecke eigenvalues. The claim $\text{ITF}(M_{\text{CKM}}) = \mathbb{Q}(\sqrt{17})$ from earlier work is corrected: the true ITF remains to be determined.

Data Availability

SnapPy computations at 150-bit precision. LMFDB data at <https://www.lmfdb.org>. Scripts at <https://github.com/drmlgentry/hyperbolic-flavor-scan>.

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